One split at a time is usually enough: empirical limits of lookahead decision tree induction

Abstract

We study whether multi-step split optimization materially improves decision trees and forests over greedy CART-style induction, once predictive performance and computational cost are evaluated jointly. Lookahead induction is theoretically attractive: a split that is weak at the current node may enable better descendant splits, so optimizing only one split at a time can be suboptimal. We implemented lookahead split selection for decision trees and forests and evaluated lookahead depths 1, 2, and 3 on 67 completed PMLB classification datasets. Depth 1 corresponds to standard greedy induction. For single trees, mean accuracy increased from 0.6931 at depth 1 to 0.7100 at depth 2 and 0.7145 at depth 3, while mean fit time increased from 0.008 s to 0.373 s and 70.90 s. For forests, mean accuracy increased from 0.7068 to 0.7125 and 0.7260, while mean fit time increased from 0.019 s to 0.694 s and 96.19 s. These results show that multi-step split optimization can improve predictive performance, but in this benchmark the gains are thin, inconsistent, and very expensive. Greedy induction remains surprisingly competitive when accuracy and cost are evaluated together.

1. Introduction

Decision trees are usually induced one node at a time. In CART-style algorithms, each internal node chooses the split that most improves a local impurity criterion, without jointly optimizing the downstream subtree. This greedy strategy is computationally convenient and remains central to single trees and random forests, but it is not globally optimal. A locally attractive split may lead to a poor subtree, and a locally unimpressive split may create children that are easy to separate.

Lookahead induction directly addresses this myopia. Rather than scoring only the immediate children of a candidate split, an n-sighted tree scores the best subtree available over n generations. For a node containing samples S and impurity I, a depth-n split may be viewed as maximizing the reduction

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I(S) - sum_{leaves L at depth n} |L| / |S| * I(L),
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after optimizing the descendant splits below the candidate root split. The usual greedy tree is the special case $n = 1$.

This observation gives a clear theoretical motivation for lookahead: there are classification problems where a one-step split score is misleading, while a two- or three-step objective can identify a better root split. The practical question is different. Does this additional optimization materially improve ordinary decision trees and forests once the computation required to search over descendant splits is included?

This short paper answers that question empirically. We ask whether multi-step split optimization materially improves decision trees and forests over greedy CART-style induction, once predictive performance and computational cost are evaluated jointly.

2. Methods

We compare lookahead depths 1, 2, and 3 for a single decision tree and for a random forest whose individual trees use the same lookahead rule. All models use axis-aligned splits. Depth 1 is the greedy baseline; depths 2 and 3 optimize the split objective over one and two additional generations of descendants.

The benchmark uses PMLB classification datasets with a fixed experimental protocol: stratified train-test split, test_size = 0.25, random_state = 0, max_depth = 3, max_features = None, and no explicit max_split_candidates restriction. Datasets with number of rows times number of predictors above 25000 were skipped after earlier timing experiments showed that larger problems were impractical for depth-3 lookahead. Datasets that failed during loading or fitting were skipped. If any model fit exceeded 30 minutes, the dataset was excluded from the completed-dataset analysis.

The final completed benchmark contains 67 datasets and 402 model fits:

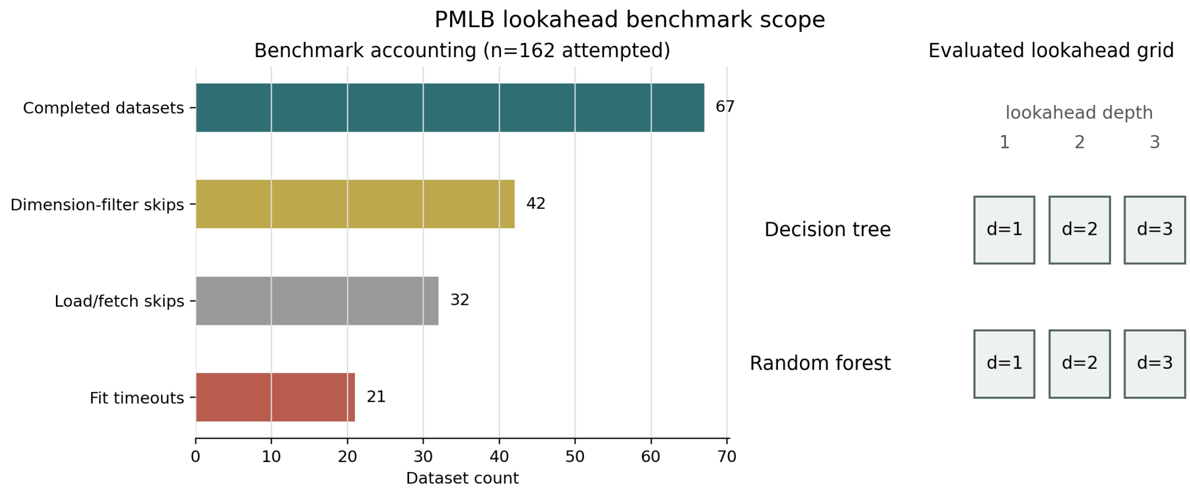


Figure 1: Benchmark accounting and evaluated grid

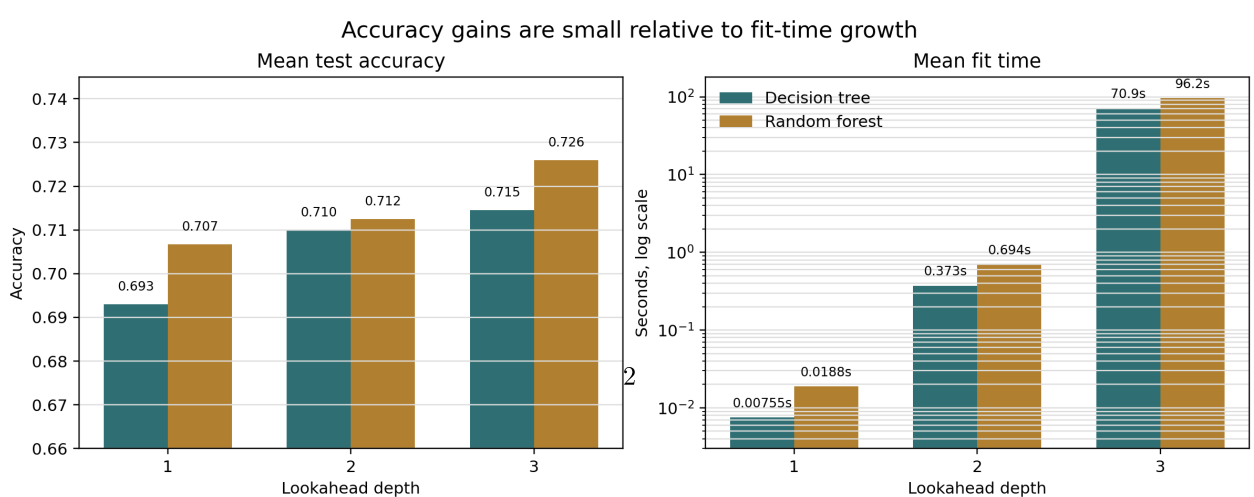
Figure 1. Benchmark scope for the PMLB lookahead experiments. The completed analysis includes 67 datasets. Additional attempted datasets were skipped by the dimension filter, by load or split failures, or by fit-time timeouts.

3. Results

Table 1 summarizes predictive accuracy and fit time. The first-order result is simple: deeper lookahead improves mean accuracy, but the improvement is small relative to the increase in fit time.

Estimator	Lookahead	Mean accuracy	Median accuracy	Mean fit time (s)	Median fit time (s)
Decision tree	1	0.6931	0.7554	0.0076	0.0043
Decision tree	2	0.7100	0.7222	0.3727	0.0547
Decision tree	3	0.7145	0.7333	70.9025	2.1834
Random forest	1	0.7068	0.7500	0.0188	0.0116
Random forest	2	0.7125	0.7407	0.6940	0.1208
Random forest	3	0.7260	0.7612	96.1864	4.0572

Table 1. Aggregate performance over the 67 completed PMLB classification datasets.



The gains are also not consistent enough to make depth 3 an obvious default. Counting ties as wins, depth 1 was tied for the best tree accuracy on 30 datasets, depth 2 on 31 datasets, and depth 3 on 37 datasets. For forests, the corresponding counts were 36, 29, and 29. Thus, deeper lookahead can help, but greedy induction remains competitive across many datasets.

4. Discussion

These experiments support a restrained conclusion. Multi-step split optimization can improve decision trees and forests, and the theoretical motivation is real. Greedy induction is myopic, and lookahead can recover splits that a one-step impurity criterion undervalues. However, under this benchmark, the empirical advantage is usually modest and comes with a large computational cost.

This matters because speed is part of the appeal of decision trees and forests. They are valued not only for interpretability and predictive performance, but also for fast fitting, simple tuning, and robustness across many tabular problems. A method that improves accuracy by one or two percentage points while increasing fit time by orders of magnitude must be justified by a specific problem need. As a routine replacement for greedy induction, depth-2 and depth-3 lookahead do not clearly clear that bar.

The forest result is particularly instructive. Random forests already reduce some instability of greedy trees by aggregating many fitted trees. In these experiments, lookahead still improves the average forest accuracy at depth 3, but the computational price remains severe. This suggests that ensembling does not make deep lookahead free; it multiplies the search cost across trees.

5. Limitations

The analysis is intentionally narrow. It evaluates a single implementation, a fixed train-test split, shallow trees with max_depth = 3, classification datasets from PMLB, and lookahead depths no larger than 3. Different pruning rules, candidate-split restrictions, caching strategies, parallelization, hardware, or hyperparameters may change the tradeoff.

The completed-dataset averages should also be interpreted with the skip accounting in mind. The benchmark excluded 42 datasets by the dimension filter, 32 datasets because of load or split failures, and 21 datasets because at least one fit timed out. These exclusions do not invalidate the completed results, but they do emphasize the practical fragility of deeper lookahead.

Finally, the present benchmark reports aggregate performance. A journal-length version should preserve and analyze the per-dataset effects, including wins, losses, uncertainty intervals, memory use, and sensitivity to forest size and tree depth.

6. Conclusion

Lookahead split optimization is a principled correction to the local myopia of greedy decision-tree induction. In the PMLB benchmark studied here, it does improve mean accuracy for both trees and forests. The improvement, however, is thin and not consistent, while the computational cost is huge. For routine tabular modeling with shallow trees, one split at a time is usually enough. Deeper lookahead appears better suited to targeted settings where computation is cheap, tree depth is small, and the expected gain is problem-specific.

References

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